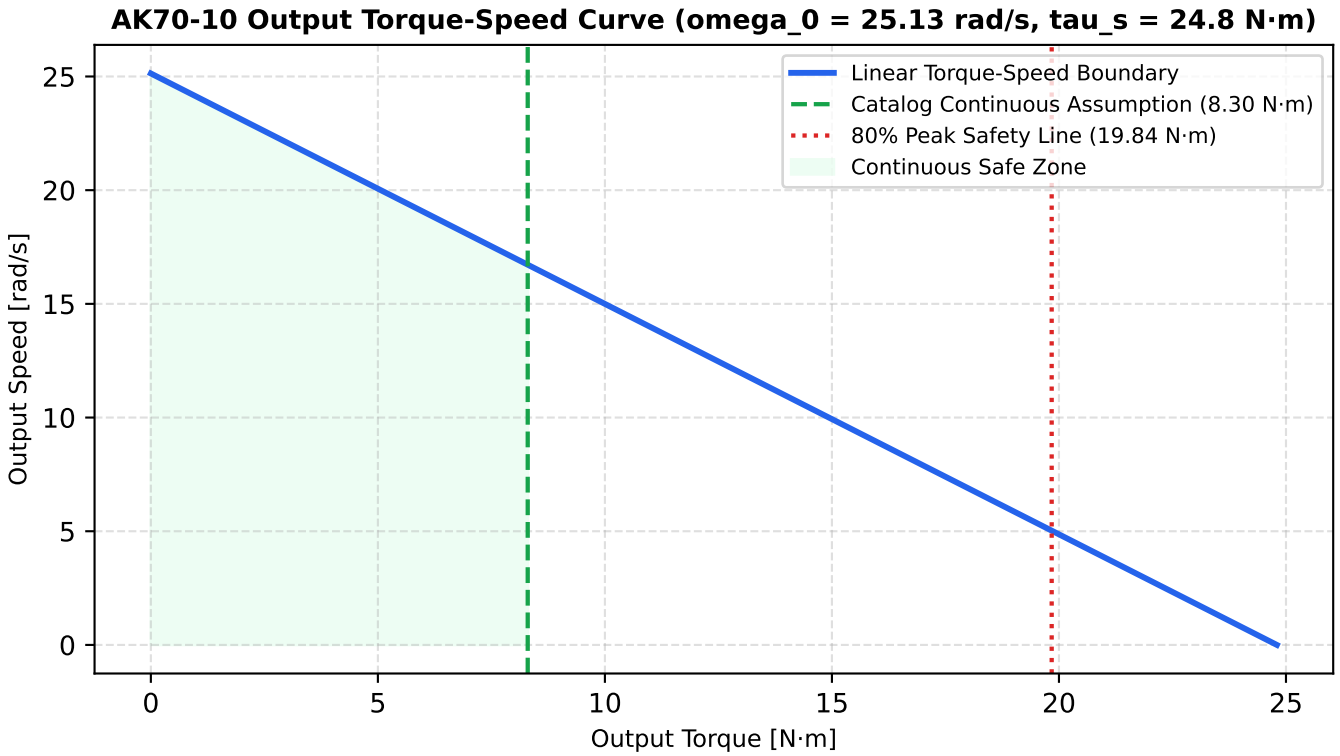
# Part Number: AK70-10



## 1. Electrical & Mechanical Specifications (Standard Reference)

| Parameter | Symbol | Specification Value | Test Condition / Notes |
|---|---|---|---|
| Part Number | ID | AK70-10 | AK-Series QDD |
| Manufacturer | Vendor | CubeMars | Outer-Rotor BLDC QDD |
| Rated Input Voltage | V_rated | 24.0 V | Nominal DC Bus |
| Stall / Peak Torque | tau_stall | 24.8 N.m peak torque | At 24.0 V |
| Continuous Rated Torque | tau_cont | 8.3 N.m rated torque | 25 degC ambient, continuous duty |
| No-Load Speed | omega_0 | 240 rpm (at 24V KV100) | Unloaded output shaft |
| Actuator Mass (Weight) | m_act | 521 g | Includes housing & internal gearhead |
| Internal Reduction | i_int | 10 : 1 Planetary QDD | Integrated gear stage |
| Envelope Dimensions | W x H x D | 80.0 x 50.0 x 80.0 mm | Outer housing envelope |



2. Mechanical Mounting & Frame Integration
   - Housing Envelope: 80.0 mm x 50.0 mm x 80.0 mm
   - Catalog Mass (unverified): 521 g (0.5210 kg)
   - Mounting Pattern: Not verified; obtain manufacturer flange/shaft drawings.

3. Thermal & Duty-Cycle Notes
   - Catalog continuous torque is an input assumption until its manufacturer rating is check
   - The 0.33x stall estimate, where used, is a heuristic rather than a measured thermal rat
   - The linear torque-speed envelope is a screening model, not a measured performance curve

4. Project Reference Index
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